

2 . Sequential Model, Queuing Theory and Related Problems

Algorithm of Sequencing Problem (n jobs, 2 machines)

Johnson's Rule (Steps):

1. **List all jobs** with processing time on Machine 1 (M1) and Machine 2 (M2).
2. **Find the smallest processing time** among all jobs.
3. If the smallest time is:
 - On **M1** → place that job at the **beginning** of the sequence
 - On **M2** → place that job at the **end** of the sequence
4. **Remove that job** from the list.
5. **Repeat** steps 2–4 until all jobs are scheduled.
6. The obtained order is the **optimal sequence**.

Algorithm of Sequencing Problem for 3 Machines (Special Case)

Convert into 2-machine problem:

- $A = M1 + M2$
- $B = M2 + M3$

Then apply **Johnson's Rule** on A and B.

Total Elapsed Time (Makespan) – Steps

1. Draw a **Gantt Chart (time table)**.
2. For first job:
 - Start on M1 at time 0
 - Move sequentially to M2, M3
3. For next jobs:
 - Start on M1 immediately after previous job finishes
 - On M2/M3 → start only when:
 - job is completed on previous machine AND
 - machine is free
4. Continue till last job completes.
5. **Total elapsed time = completion time of last job on last machine**

Idle Time Calculation

Formula:

Idle Time = Total elapsed time – Total processing time on that machine

Steps:

1. Find **total elapsed time (T)**
2. Find total processing time on each machine:
 - M1 total = sum of all M1 times
 - M2 total = sum of all M2 times
 - M3 total = sum of all M3 times
3. Idle time:
 - Idle M1 = T – (Total M1 time)
 - Idle M2 = T – (Total M2 time)
 - Idle M3 = T – (Total M3 time)

Important formulae of Queuing Theory (M/M/1 model):

Basic Terms

- λ = Arrival rate (customers per unit time)
- μ = Service rate (customers per unit time)
- ρ = Utilization factor

1. Utilization Factor/ Traffic Intensity

$$\rho = \lambda / \mu$$

Condition:

System is stable if $\rho < 1$

2. Probability of Zero Customers (Idle System)

$$P_0 = 1 - \rho = 1 - \lambda / \mu$$

3. Probability of n Customers in System

$$P_n = (1 - \rho) \times \rho^n = P_0 \cdot (\lambda / \mu) \cdot n$$

4. Average Number of Customers in System (L)

$$L = \lambda / (\mu - \lambda)$$

5. Average Number of Customers in Queue (Lq)

$$L_q = \lambda^2 / [\mu(\mu - \lambda)]$$

6. Average Waiting Time in System (Ws)

$$W = 1 / (\mu - \lambda)$$

7. Average Waiting Time in Queue (Wq)

$$W_q = \lambda / [\mu (\mu - \lambda)]$$

8. Relationship (Little's Formula)

$$L = \lambda W,$$

$$L_q = \lambda W_q$$

9. Probability that Customer has to Wait

$$P(\text{wait}) = \rho$$

10. Probability of More than n Customers

$$P(N > n) = \rho^{n+1}$$

11. Idle Time

$$P(\text{idle}) = 1 - \rho$$